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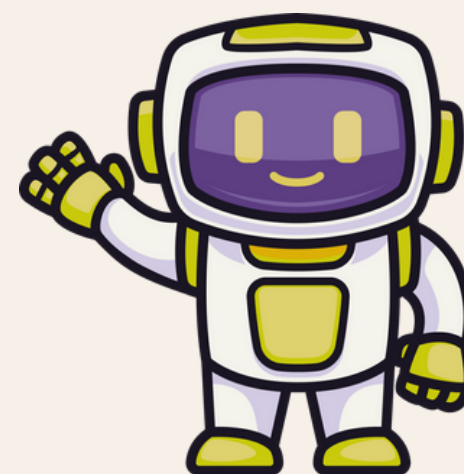


What's new in **Angular 21**



Build with AI

<https://angular.dev/ai>



- Your resource for everything you need to build AI-powered applications with Angular.
- This guide demonstrates how you can use **Genkit**, Firebase AI Logic, and the Gemini API to infuse your Angular apps with AI today.



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Experimental Signal Forms

Reactive Forms

- Explicitly define FormGroup and FormControl instances using FormBuilder or new operators.
- Uses [formGroup] and FormControlName directives.
- Validators are passed as an array when creating FormControls.
- Access state (e.g., form.valid) via properties updated by change detection.

Signal Forms

- Define a plain object as a signal() and pass it to the form() function.
- Uses the **[field] directive** for two-way binding.
- Validation rules are defined centrally in a schema function passed to form(), with built-in error messages.
- Access state (e.g., form.valid()) by **calling the field/form as a function (signals)**.



Lets understand with Login form example

```
// 📌 signal to hold the form credentials
private readonly credentials = signal({
  login: '',
  password: ''
});
// 📌 form created from the credentials signal (type is `FieldTree`)
protected readonly loginForm = form(this.credentials);
```

```
<form (submit)="authenticate($event)">
  <label for="login">Login</label>
  <input id="login" [field]="loginForm.login" />
  <label for="password">Password</label>
  <input id="password" [field]="loginForm.password" />
  <button type="submit">Log in</button>
</form>
```

field directive





```
protected async authenticate(event: SubmitEvent) {  
  event.preventDefault();  
  await submit(this.loginForm, form => {  
    const { login, password } = form().value();  
    return this.userService.authenticate(login, password);  
  });  
}
```



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Bye Bye Zonejs



Zoneless Change Detection by Default: New applications no longer rely on Zone.js for change detection, making the framework more aligned with modern reactive principles, improving performance, and simplifying debugging.



HttpClient by Default

The HttpClient is now provided in the root injector by default, meaning fewer imports are needed in new projects.



Vitest as the Default Test Runner

Vitest replaces Karma/Jasmine as the default unit testing framework for new projects, offering faster test execution and a better developer experience



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Developer Experience (DX) and Tooling

AI-Powered Developer Tools (MCP Server)

The Model Context Protocol (MCP) developer server integrates AI capabilities into the CLI, providing smart migration guides, code suggestions, and the ability to generate components and services from natural language specifications.

Improved Accessibility (Angular ARIA)

A new experimental @angular/aria package provides headless and accessible directives for common UI patterns (e.g., Accordion, ComboBox, Tabs), simplifying the process of building accessible components.



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